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Assignment - 4

SECTION-A

Ans-1 SRAM vs DRAM → SRAM uses flip flops to store data making it faster, expensive and power consuming. It is used in cache memory. DRAM uses capacitors to store data, needs refreshing, slower but cheaper.

Ans-2 Programs tend to access a small portion of memory repeatedly.

Type
- → Temporal locality - recently accessed data
- → Spatial locality - nearby locations are access

Ans-3 Cache memory stores frequently used data / instructions closer to CPU. It reduces avg memory access time & improves performance.

Ans-4 Cache Indexing
- → Direct mapped cache
- → Set associative cache
- → Fully associative cache.

Ans-5 Larger cache block size increases spatial locality [fewer misses] but increases miss penalty and may load unnecessary data.

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Ans-6 Associativity is the number of locations where a memory block can be placed in order. High associativity reduces conflict misses but increases hardware cost.

Ans-7 Larger cache size reduces miss rate and improves performance but increases cost, power & access latency.

Ans-8

(i) Write-through : Data is written to both cache & main memory at same time.

(ii) Write-Back : Data is written to cache first & main memory is only updated when block is replaced.

Ans-9 AMAT = Hit time + (Miss Rate × Miss Penalty)

Hit time → time to access cache

Miss Rate → probability of miss

Miss Penalty → time to fetch from main memory

Ans-10 LRU block is replaced first. The block which has not been used for longest time is removed.

Ans-11 Memory is divided into multiple modules and accessed simultaneously in a round robin fashion. It increases memory bandwidth and improves perf-

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Ans-12 Tracks are concentric circular paths on disk surface. Sectors are sub divisions of tracks used to share data block.

Ans-13 Flash memory is a non volatile memory that retains data even when power is off. Used in SSDs, pen drives, memory cards faster than magnetic disks and has no moving parts.

Ans-14 I/O devices share the same address space as memory. Devices are accessed using normal memory instructions.

Ans-15 I/O devices have a seperate address space. They are accessed by using special I/O instructions.

Ans-16 CPU continaously checks the status of I/O device and controls the data transfer

Ans-17 Device interrupt the CPU when it is ready CPU can perform other tasks meanwhile. Better CPU Utilization.

SECTION-B

Ans-1 Data flow b/w different levels uppers levels are faster & smaller, Lower levels are slower but larger.

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Registers
↕
Cache
↕
Main Memory
↕
Secondary memory.

Ans-2

Address		
Tag (t bits)	Index (i bits)	Offset (b bits)

Cache Levels	Valid	Tag	Data Block (2^b bytes)
0			
1			
2			
⋮			
$2^i - 1$			

Ans-3 i) Write-through

CPU
↓ write
Cache
↓
Main memory

ii) Write-Back

CPU
↓ write
Cache
↕ (write on replacement)
Main memory

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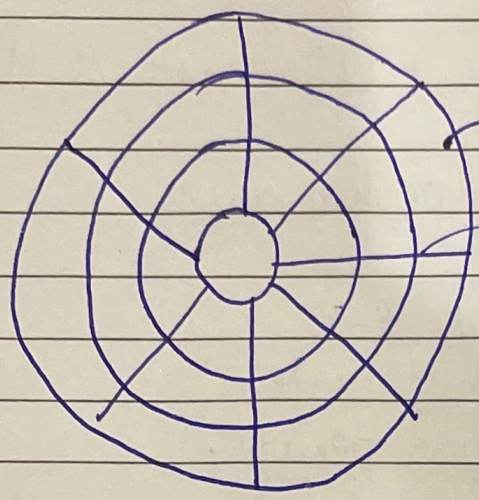
Ans-4



SECTION-C

Ans-1 Temporal locality, because the same location is accessed again & again.

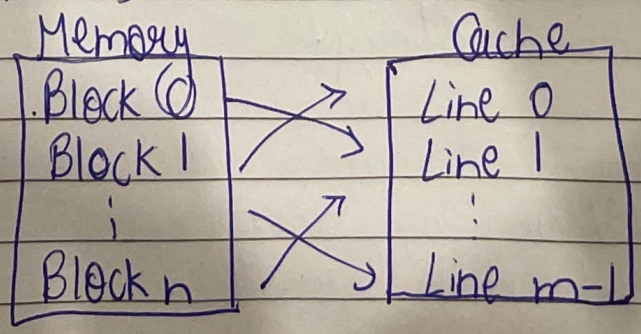
Ans-2 Larger blocks increase miss penalty (more data to be transferred) and may load unnecessary data (Cache pollution)

Ans-3 On every write, data is written to both cache & main memory

Ans-4 Direct memory access (DMA)

SECTION-D

Ans-1



Mapping: memory block is placed in cache line using index. Tag is used in index

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Ans-2 AMAT = Hit time + (Miss rate × Miss Penalty)

Factors affecting performance
- → lower hit time
- → lower miss rate
- → lower miss penalty.

These reduces AMAT & improve performance.

Ans-3

Programmed I/O	Interrupt I/O	DMA
→ CPU checks devices continuously	Device interrupts CPU when ready	DMA transfers to and from memory.
→ CPU is busy waiting	Better CPU utilization	Best efficiency

SECTION-E

Ans-1 Two assumptions while explaining cache operation
- → Fixed batch size is assumed
- → Single level cache is considered

Ans-2 Increases memory traffic because every write is updating main memory.

Ans-3 Cache mapping is most difficult because it involves address breakdown (tag, index, offset) and mapping logic conceptual and needs practice.

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